

Air vs Rail Travel Time Analysis in European Cities

Project Work Plan

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Question Refinement

Refinements to initial questions:

- Considering only cities with a population $\geq 500K$
- Only looking at direct flight connections
- Setting fixed transit times for air and rail

With this refinement, we want to answer the following questions:

- How do the travel times compare between rail and air?
- Which connections are faster by train than by air?
- Which connections could be faster by train than by air if high-speed rail was possible?
- Which are the 5 most well connected cities and the 5 least well connected?

We plan to do that by:

- Calculating the travel time estimate for both rail and air for each city pair
- Compare total journey times and identify routes where rail time $\geq$ air time
- Apply theoretical speed of 300 km/h to rail routes, recalculate travel times with high-speed assumption and identify newly competitive rail routes
- Calculate number of minimum-time connections per city and rank cities based on connectivity metrics
- Visualizing the connectivity with isochrones

Data Sources & Current Status

During our initial data collection, we realized that the travel time data statistics for air and rail travel are not accessible in a single data base. Also, data on time schedules for all Europe was as far as we researched behind paywalls. Therefore, we developed different strategies to land at realistic time estimates.

City Data

- [EUROSTAT population database](#)
- [OpenDataSoft city coordinates](#)
- Status: Initial data set created, 80 cities identified

Air Travel

- [European Airports IATA Codes](#)
- [Airport database with coordinates](#)
- [Flight Route Database](#) (7 years old)
- [European Flights Dataset](#)
- Status: Direct flight connections mapped, travel time calculated with haversine formula and average speed, transit time integrated

Rail Travel

- [Train station data from Kaggle](#)
- OpenStreetMap rail network data via OSMnx
- [Rail network visualization](#)
- Status: Initial OSMnx implementation tested, needs scaling, takes a long time to load, searching for more efficient ways currently.

Additional Resources

- [EU Transport Analysis](#): In this report, similar questions are analyzed. We looked so far especially at the methodology to find hints on how to get the right data. However, they apparently had access to a much wider base of datasets.
- [Transit time research paper](#): This paper examined transit times for airports and railway stations.

Work Distribution

In regular group meetings, we discuss general topics like the refinement of the questions and the formation of tasks to answer them. From that we split the following sub-tasks that we synchronize regularly.

- Anabel: Gathering of air travel data
- Ralf: Gathering of rail travel data, Comparative Analysis
- Wendelin: Visualization, Setting up shared environment
- Hedda: Gathering of rail travel data, Putting together presentation files and report